

Matthew Macfarlane

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EDUCATION

PhD candidate Artificial Intelligence 2021 - Present
University of Amsterdam, Netherlands

- Deep Reinforcement Learning with Search and efficient test time adaptation.
- Supervised by Herke van Hoof, within the AMLab group.

MSc in Artificial Intelligence (Distinction) 2019 - 2020
University of St Andrews, UK

- Graduated with distinction, scoring the highest average grade in the cohort.
- Thesis: Application of Reinforcement Learning for Combinatorial Optimisation

BSc in Mathematics with Economics (First Class) 2014 - 2017
The London School of Economics, UK

- First class honors with highest marks in the cohort for Algebra and its Applications.

EXPERIENCE

Research Scientist (Lila Sciences) 2025
• Working on Open endedness.

Research Intern (Microsoft Research, Montreal) 2025
• Working on reasoning in LLMs, with Alessandro Sordoni.

Visiting Researcher (University of Alberta, Canada) 2024 - 2025
• Working with Levi Lelis on learning neural languages as discrete representations for meta-learning.

Deep Reinforcement Learning Intern (InstaDeep, Amsterdam) 2023 - 2024
• Algorithms combining probabilistic inference and amortization for control.
• Scalable reinforcement learning algorithms and environments using JAX.

Artificial Intelligence Intern (Techspert.io, Cambridge, UK) 2020
• Applied reinforcement learning to large-scale web scraping.

Multi-Asset Quantitative Analyst (Aberdeen Standard Investments, UK) 2017 - 2019
• Conducted quantitative analysis using time-series econometrics and machine-learning methods for multi-asset investment funds.

Part-time Live Sports Trader (Kambi, UK) 2016 - 2017
• Managed live in-play sports betting odds for football matches.

Project Assistant (University of Aberdeen, UK) 2012 - 2014
• Data collection for Ankylosing Spondylitis and statistical analysis using SPSS.

PAPERS

Searching Latent Program Spaces 2025
• NeurIPS 2025 main track Spotlight paper, first author
• ICML 2025 PRAL Workshop Paper and Oral Presentation.
• ARC AGI 2024 Paper Competition, 3rd Prize Award. Available on arXiv: [arXiv:2411.08706](https://arxiv.org/abs/2411.08706).
• Featured on Machine Learning Street Talk: [YouTube link](#).
• GitHub repository: github.com/clement-bonnet/lpn.

- Instilling Parallel Reasoning into Language Models** 2025
- ICML 2025 2nd AI for Math Workshop paper, first author.
- SPO: Sequential Monte Carlo Policy Optimisation** 2024
- NeurIPS 2024 main track, first author. *Available on arXiv: 2402.07963.*
 - Featured in *Reinforcement Learning: A Comprehensive Overview* (Murphy 2024)
- Jumanji: A Diverse Suite of Scalable Reinforcement Learning Environments** 2024
- ICLR 2024 main track. *Available on arXiv: 2306.09884.*
 - GitHub repository: <https://github.com/instadeepai/jumanji>.
- Graph Neural Networks with Policy Gradients for Tree Search** 2022
- NeurIPS 2022 Deep Reinforcement Learning Workshop, first author.
 - *OpenReview Link: 0Hm6VYaAiRP.*

Google Scholar: <https://scholar.google.com/citations?user=fxbRMT4AAAAJ&hl=en>.

OPEN SOURCE CONTRIBUTIONS

Flashbax

- Co-author of an open-source repository for high-speed implementation of replay buffers in JAX for Reinforcement Learning. Utilized by Deepmind.
- GitHub repository: <https://github.com/instadeepai/flashbax>.
- Stars: 227

Jumanji RL Environments

- A diverse suite of scalable reinforcement learning environments written in JAX.
- GitHub repository: <https://github.com/instadeepai/jumanji>.
- Stars: 701

Latent Program Networks

- An open-source implementation on self-adapting latent program neural networks.
- GitHub repository: <https://github.com/clement-bonnet/lpn>.
- Stars: 69

Stoix RL Library

- An open-source reinforcement learning research library.
- GitHub repository: <https://github.com/EdanToledo/Stoix>.
- Stars: 287

SKILLS

Programming

- **Python (Advanced):** Expertise in designing and implementing machine learning algorithms, using a range of libraries such as JAX and PyTorch.
- **JAX:** Extensive experience in building high-performance reinforcement learning environments and algorithms.
- **Cluster Computing:** Proficient in managing GPU and TPU clusters / pods (e.g., TPU-v4 - (8-256) for large-scale machine learning training.

CERTIFICATIONS & AWARDS

- UK Mathematics Trust Challenge, Gold Medal (2009-2014)
- CFA Level 2